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"Vegetation and Energy"

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ALGORITHM THEORETICAL BASIS DOCUMENT

LAND SURFACE TEMPERATURE - LST V3.0

Issue I1.10

Organization name of lead contractor for this deliverable: IPMA

Book Captain:	Sofia Ermida
Contributing Authors:	Emanuel Dutra
	Isabel Trigo
	Francisco Lopes
	João Martins

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List of Acronyms

ABI	Advanced Baseline Imager
AD	Applicable Document
AHI	Advanced Himawari Imager
ASTER	Advanced Spaceborne Thermal Emission and Reflection Radiometer
ATBD	Algorithm Theoretical Basis Document
AVHRR	Advanced Very High-Resolution Radiometer
CAL/VAL	Calibration and Validation
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
ECMWF	European Centre for Medium-Range Weather Forecasts
EUMETCast	EUMETSAT's broadcast system for environmental data
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FVC	Fraction of Vegetation Cover
FCOVER	CGLOPS Fraction of Vegetation Cover product
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEO	Geostationary satellites
GOES	Geostationary Operational Environmental Satellite
GSW	Generalized Split-Window
HW/SW	Hardware / Software
IPMA	Instituto Português do Mar e da Atmosfera
IGBP	International Geosphere-Biosphere Program
IMS	Snow and Ice Mapping System
INSPIRE	Infrastructure for Spatial Information in Europe
IODC	Indian Ocean Data Coverage
IR	InfraRed
JHU	John Hopkins University
JMA	Japan Meteorological Agency
JRC	Joint Research Centre
LPV	Land Product Validation

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LSA-SAF	EUMETSAT Satellite Application Facility on Land Surface Analysis
LST	Land Surface Temperature
LUT	Look-Up-Tables
MODIS	Moderate Resolution Imaging Spectroradiometer
MODTRAN6	MODerate spectral resolution atmospheric TRANSsmittance
MSG	Meteosat Second Generation
MTG	Meteosat Third Generation
NetCDF	Network Common Data Form
NHEMI	Northern Hemisphere
NOAA	National Oceanic and Atmospheric Administration
NRT	Near Real Time
NWC SAF	EUMETSAT SAF on Nowcasting and Very Short-Range Forecasting
NWP SAF	EUMETSAT SAF on Numerical Weather Prediction
OLCI	Ocean and Land Colour Instrument
PUM	Product User Manual
QAR	Quality Assessment Report
RMSD	Root Mean Square Difference
SAF	Satellite Application Facility
SCE	Snow Cover Extent
SEVIRI	Spinning Enhanced Visible Infra-Red Imager
SZA	Satellite (viewing) Zenith Angle
T2m	Temperature at 2 meters height
TCWV	Total Column Water Vapour
TIGR	Thermodynamic Initial Guess Retrieval
TIR	Thermal InfraRed
TOA	Top-of-Atmosphere
TSkin	Skin temperature
UCSB	University of California Santa Barbara
USNIC	United States National Ice Center
VR	Validation Report

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This Algorithm Theoretical Basis Document (ATBD) describes the scientific background and the procedures used to calculate the global Land Surface Temperature (LST) obtained from a constellation of geostationary (GEO) satellite missions: Meteosat Second Generation (MSG) – 0° service and Indian Ocean Data Coverage (IODC) service; Geostationary Operational Environmental Satellite (GOES); and Himawari. These platforms are operated by different satellite agencies, but the data are made available in near-real time (NRT) via the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) dissemination system, EUMETCast.

LST is the radiative skin temperature of land surface, as measured in the direction of the remote sensor, corrected for surface emissivity. This document describes the LST retrieval methodology that is based on semi-empirical functions that relate LST to top-of-atmosphere brightness temperatures in thermal window channels (around 11 μm and 12 μm). The assumptions and physics underlying this methodology, as well as the uncertainties of LST estimates are discussed. The formulation is trained using a dataset of radiative transfer simulations for a wide range of atmospheric and surface conditions. The performance of the algorithm is assessed by comparing its output against an independent set of simulations.

Surface emissivity is also a critical input of the processing chain. Within this product, it is estimated using the vegetation cover method, which weights bare ground and vegetation emissivities by the fraction of vegetation cover of each pixel. Vegetation and bare ground emissivities are prescribed for each land cover class, based on information derived from comprehensive spectral libraries, while the fraction of vegetation cover is based on the CLMS Fraction of Vegetation Cover (FCOVER) 300m product. Surface emissivity is also affected by the presence of snow, which is here accounted for using the CLMS Snow Cover Extent (SCE) 1km product, complemented by the Snow and Ice Mapping System (IMS) Daily Northern Hemisphere Snow and Ice Analysis from the United States National Ice Center (USNIC) over cloud gaps.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document includes the theoretical basis of the LST V3.0 retrieval methodology and details the physics background of the different algorithms, as well as expected accuracy, which are associated to the algorithms' uncertainties and to inputs errors, and limitations of the remote sensing data.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements.
- Chapter 3 describes the methodologies that shall be used for LST estimation.
- Chapter 4 describes the known limitations, summarizing some quality assessment results.
- Chapter 5 presents the output LST product.
- Chapter 6 identifies the risks and the mitigation measures.
- Chapter 7 lists the references.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at: https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_FCOVER300m-V1.1	Algorithm Theoretical Basis Document of the FCOVER 300m Version 1.1 product
CGLOPS1_ATBD_FCOVER300m-V2	Algorithm Theoretical Basis Document of the FCOVER 300m Version 2.0 product

CGLOPS2_ATBD_SCE-NHEMI-1km_V1	Algorithm Theoretical Basis Document of the Northern Hemisphere Snow Cover Extent 1km V1.0 product
CGLOPS2_ATBD_SCE-GLOBE-1km_V1	Algorithm Theoretical Basis Document of the Global Snow Cover Extent 1km V1.0 product
CGLOPS2_QAR_SCE-NHEMI-1km_V1	Quality Assessment Report of the Northern Hemisphere Snow Cover Extent 1km Version 1.0 product

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_LST-V3.0	Product User Manual summarizing all information about the LST Version 3.0 product
CGLOPS1_VR_LST-LST-V3.0	Validation report of the LST Version 3.0 product

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements
Available at https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1	
IMAGINES_RP2.1_ATBD-LAI300m	Algorithm Theoretical Basis Document of the FCOVER 300m Version 1.0 product
Available at: https://land.copernicus.eu/en/technical-library	
SAF/LAND/IM/ATBD_MLST/1.2	Algorithm Theoretical Basis Document for MSG Land Surface Temperature (LSA-001, LSA-050), version 1.2, Issue 2
Available at: https://nextcloud.lsasvcs.ipma.pt/s/Dd6ga76oG7Lqz5Q	
NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0	Scientific and Validation report for the Cloud Product Processors of the NWC/GEO, Issue 1, Rev. 0, 2016

Available at:
http://www.nwcsaf.org/AemetWebContents/ScientificDocumentation/Documentation/GEO/v2016/NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0.pdf

JMA, 2017 Himawari-8/9 - Himawari Standard Data User's Guide, v1.3. Tokyo

Available at:
https://www.data.jma.go.jp/mscweb/en/himawari89/space_segment/hsd_sample/HS_D_users_guide_en_v13.pdf

NOAA, 2019 GOES R Series Product Definition and Users' Guide. 416-R-PUG-L1B-0347 Vol 3 Rev.2.2

Available online at: https://www.goes-r.gov/users/docs/PUG-L1b-vol3.pdf

U.S. National Ice Center, 2008 *IMS Daily Northern Hemisphere Snow and Ice Analysis at 1 km, 4 km, and 24 km Resolutions*. (G02156, Version 1). Boulder, Colorado USA. National Snow and Ice Data Center. https://doi.org/10.7265/N52R3PMC

Available at: https://nsidc.org/data/g02156/versions/1

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the LST product are summarized below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global

LST – LAND SURFACE TEMPERATURE	
The Land Surface Temperature (LST) is the radiative skin temperature of the land surface, as measured in the direction of the remote sensor [K].	
Spatial resolution	5 km
Time span:	Hourly and 10-day period
Timeliness:	Hourly: within 4 hours; within 2 days (optimally 1 day) after the end of 10-day period.
Uncertainty (2 sigma)	Threshold: 2 K, Goal: 1 K
Stability (per decade)	Threshold: 0.3 K, Goal: 0.1 K

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets shall be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

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3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

LST is estimated from Top-of-Atmosphere (TOA) brightness temperatures of atmospheric window channels within the infrared range. The used *split-window* methodology makes use of two adjacent window channels within the thermal infrared range (T_{TIR1} and T_{TIR2}), where LST is expressed as a regression function of TOA brightness temperatures. To minimize LST uncertainties, the algorithm is trained for different classes of satellite viewing angle and atmosphere water vapour content.

In the current version of CGLOPS LST, the global LST is derived using measurements from the Advanced Himawari Imager (AHI) onboard Himawari-8 and Himawari-9, the Advanced Baseline Imager (ABI) onboard GOES-16 and GOES-19, and the Spinning Enhanced Visible Infra-Red Imager (SEVIRI) onboard the Meteosat Second Generation (MSG) platforms in charge of the 0º and Indian Ocean Data Coverage (IODC) missions. The product includes data since January 1st 2018, ensuring coverage from all the above-mentioned sensors.

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Outline

The LST product is generated every hour, through the following steps:

Step 1: Generation of LST product for each GEO satellite disk.

Step 1.1: Gathering of input data in the same resolution and projection as the satellite imagery;

Step 1.2: LST production is performed for all land pixels;

Step 1.2.1: the LST is calculated with the Generalized Split Window (GSW) algorithm (section 3.2.3); where surface emissivity is first estimated considering the Land Cover, Fraction of Vegetation Cover and Snow Cover of each pixel.

Step 1.2.2: the uncertainties associated to the LST retrievals are calculated and set up as quality flag;

Step 2: The LST (and the additional layers) for the Himawari and GOES disks is merged with the LSA-SAF LST from both MSGs (0° and IODC missions) to obtain a global product.

3.2.2 Basic underlying assumptions

LST estimations from remotely sensed data are generally obtained from one or more channels within the thermal infrared atmospheric window from 8-to-13 μm (Dash et al., 2002, Li et al, 2013). Operational LST retrievals often make use of split-window algorithms (e.g., Prata, 1993; Wan and

Dozier, 1996), where LST is obtained through a semi-empirical regression of TOA brightness temperatures of two pseudo-contiguous channels, i.e., the split-window channels.

A relevant factor in the selection of the algorithm is its expected reliability for operational LST retrievals, both in terms of expected accuracy and timeliness. The latter favours the use of semi-empirical relationships between LST and TOA brightness temperatures, which are computationally efficient and free of the convergence problems of direct emissivity and temperature retrieval methods (e.g., Faysash and Smith, 1999; Hulley and Hook, 2011) associated to the non-linearity of the inverse problem in remote sensing (e.g., Rodgers, 2000).

Ultimately, we want to produce a global LST field, which corresponds to the fusion of LST fields generated from four GEO satellites: 1) GOES East, covering the Americas, 2) MSG, covering most of Europe and Africa, 3) MSG covering the Indian Ocean, the Middle East and also viewing large parts of both Europe and Africa from the East, and 4) Himawari covering Eastern Asia and Oceania. The design of the algorithms applied to each individual sensor needs to take into account the channels described in Table 1.

Table 1: Satellite channels used for LST estimation using the GSW algorithm.

	MSG/SEVIRI	GOES/ABI	Himawari/AHI
TIR1	10.0 – 11.5 µm	10.8 – 11.6µm	10.8 – 11.6 µm
TIR2	11.2 – 12.8 µm	11.8 – 12.8µm	11.7 - 13.1 µm

Both ABI onboard GOES-16 (and following satellites of the same series) and AHI onboard Himawari-8 (and following satellites) provide three channels in the split-window region. The given combination of channels was chosen to achieve the best correspondence with SEVIRI/MSG and therefore facilitate product harmonization.

3.2.3 Generalized Split-Window Algorithm

3.2.3.1 Input data

The GSW algorithm provides estimation of LST using measurements from two adjacent channels within the infrared atmospheric window, T_{TIR1} and T_{TIR2} (Table 1).

All inputs are re-gridded (both in time and space, when needed) for each pixel in the original satellite geostationary projection, so that all algorithm inputs are in a common grid. In the case of GOES-R, the radiance quality flag is used to ensure that only pixels with "good" quality are used because pixels with less-than-ideal quality can cause outliers and unrealistic LSTs.

The GSW requires the following inputs:

- Land Cover database – provided by the International Geosphere-Biosphere Program (IGBP) database (Belward, 1996). All valid pixels of the input dataset falling within each pixel of the satellite projection are selected, and the land cover type attributed to the pixel corresponds to the most frequent class.
- Fraction of Vegetation Cover for
    - SEVIRI daily pixels – estimated by the EUMETSAT Satellite Application Facility on Land Surface Analysis (LSA-SAF) from surface reflectance data (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011)
    - Other sensors – Copernicus Global Land Service FCOVER 300m product (https://land.copernicus.eu/en/products/vegetation?tab=vegetation_properties);
- Look-Up-Tables (LUT) for the bare soil and vegetation emissivities, for each land cover class, and for each TIR channel of each sensor.
- Snow fraction, derived from CLMS Snow Cover Extent 1 km product [CGLOPS2_ATBD_SCE-NHEMI-1km_V1, CGLOPS2_ATBD_SCE-GLOBE-1km_V1]); cloud gaps in the CLMS Snow Cover product are filled in with the IMS Daily Northern Hemisphere Snow and Ice Analysis 4 km product.
- Cloud mask obtained from the Nowcasting and Very Short-Range Forecasting (NWC) SAF software.
- Total Column Water Vapor (TCWV), which is obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) operational forecasts, retrieved with an hourly frequency and at 9 km resolution. Forecast fields are bilinearly interpolated to the satellite grid for the neighbouring time steps, and then linearly interpolated in time to the acquisition time of each pixel.

MSG-based LSTs (derived over the 0° nominal disk and IODC) are processed by the LSA-SAF, being then directly ingested in the CGLOPS global product, while GOES and Himawari LST fields have specific CGLOPS processing chains. The LSA-SAF LST uses the LSA-SAF Fraction of Vegetation Cover (FVC) product (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011), derived in the same processing chain and from the same sensor (SEVIRI) on a daily basis. Since the LSA-SAF FVC is limited to the MSG disks, LST retrievals from GOES and Himawari missions make use of the CGLOPS FCOVER 300m, which corresponds to the same variable (i.e., fraction of green vegetation per pixel).

Annex 1 lists the changes in input data used in the reprocessing and in the near-real time processing.

3.2.3.2 LST retrieval

The GSW algorithm, which is used for example at LSA-SAF in their LST products (Trigo et al., 2008b), uses a formulation similar to the one proposed by Wan and Dozier (1996) for the Advanced Very High-Resolution Radiometer (AVHRR) and the Moderate Resolution Imaging Spectroradiometer (MODIS). In this algorithm, LST is as a function of TOA brightness temperatures of each sensor split-window channels (T_{TIR1} and T_{TIR2}, respectively) – see Table 1):

$$LST = \left(A_1 + A_2\frac{1-\epsilon}{\epsilon} + A_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{TIR1}+T_{TIR2}}{2} + \left(B_1 + B_2\frac{1-\epsilon}{\epsilon} + B_3\frac{\Delta\epsilon}{\epsilon^2}\right)\frac{T_{TIR1}-T_{TIR2}}{2} + C + \Delta LST \quad (1)$$

where ε is the average of the two channels surface emissivities, Δε their difference (ε_{TIR1} - ε_{TIR2}), while A_j, B_j, (j = 1,2,3) and C are the GSW coefficients obtained by fitting equation ((1) to a calibration dataset for different classes of water vapour, TCWV, and satellite or viewing zenith angle (SZA), and ΔLST is the model error. The calibration and verification of the model imply the realization of a set of radiative transfer simulations of TOA brightness temperature values for a wide range of surface, atmospheric and geometric conditions, which are then separated into a calibration dataset and a verification dataset (as described below). The model error, ΔLST, may be assessed by comparing retrieved LST values to the "true" values, which were used to force the radiative transfer simulations in the verification database (which is similar, but independent of the database used to calibrate the algorithm). Note that those comparisons only assess the algorithm performance, neglecting all the input uncertainties and their propagation into the final LST estimate. The GSW algorithm is applied to clear sky pixels only (Freitas et al., 2012), and cloud are identified using the NWC-SAF software (NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0). The pixel emissivity for channel i is estimated as a function of the FVC according to the equation (Becker and Choudhury, 1988; Caselles *et al.*, 1997):

$$\boldsymbol{\epsilon_i = FVC \times \epsilon_{veg} + (1 - FVC) \times \epsilon_{BS}} \quad (2)$$

where ε_{BS} and ε_{Veg} represent the emissivities of bare soil and vegetation, respectively. The values of the emissivities (ε_{BS} and ε_{Veg}) and their uncertainties are obtained for each sensor and for each channel using a spectral emissivity library of surface materials and convolving each individual spectrum with the channel response function and then averaging for all materials within a given land cover type (Peres and DaCamara, 2005; Trigo *et al.*, 2008b). The spectral emissivity library is composed of data obtained from the Johns Hopkins University (JHU) (Salisbury and D'Aria, 1992) and Jet Propulsion Laboratory spectral libraries included in the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) spectral library, as well as from the MODIS/University of California—Santa Barbara (UCSB) spectral library. The aggregation of spectra in the different land cover classes is detailed in Peres and DaCamara (2005). The uncertainty here is the absolute uncertainty, i.e. the

largest absolute deviation from the mean emissivity within each class. Each spectrum in the spectral database is convolved with the channel response function for each sensor to obtain channel emissivities for purely bare soil and vegetation types. The spectral response functions were obtained from Numerical Weather Predition (NWP) SAF (https://www.nwpsaf.eu/site/software/rttov/download/coefficients/spectral-response-functions/). These values are shown in Table 13 to Table 15 in Annex 2: Emissivities and their uncertainties.

FVC is estimated dynamically, however it is provided by different sources. The LSA-SAF missions (MSG 0° and IODC) use their own daily FVC product, based on SEVIRI observations (García-Haro *et al.*, 2005; Roujean *et al.*, 1992; Trigo *et al.*, 2011). In the cases of GOES and Himawari missions, the dekadal CLMS FCOVER 300 m product, based on PROBA-V and Ocean and Land Color Instrument (OLCI) data onboard Sentinel-3 platform, is used. It is integrated in the CGLOPS LST processing chain as follows: first, the FCOVER 300 m product is resampled to the satellite projection corresponding to each mission, and then all the valid FCOVER grid points contained in each pixel of the GEO grid are averaged to obtain the final pixel estimate. The same applies for the FCOVER product uncertainty. These are then used as inputs for the calculation of the channel emissivities (cf. eq. (2) and their uncertainties (eq. (13 below). The FCOVER NRT estimates present many missing data (namely, in the RT0 product), especially over regions with frequent cloud cover (e.g., the tropical rainforest). In those cases, a pre-computed "climatology" for each dekad was derived from the RT6 product for the period 2018-2024, which is used to fill in the gaps of the NRT product, thus ensuring that a realistic, seasonally varying value is used in any case. This "climatology" is also used as a backup in case there is a failure in the reception of the NRT FCOVER data. A similar approach is also implemented by the LSA-SAF in the generation of the MSG LST estimates, where FVC gaps are filled with a climatology.

The GOES and Himawari retrievals are further corrected to account for the presence of non-permanent snow. This was already done in the case of the MSG 0º and IODC disks, as LSA-SAF already used an internal snow mask derived from albedo measurements obtained by SEVIRI. In the cases of the GOES and Himawari retrievals, the CLMS Snow Cover Extent 1 km product is used to estimate a snow fraction for each pixel. The 1 km data provides an estimate of the fraction occupied by snow over each grid point. Because the CLMS SCE is also affected by the cloud coverage, and to obtain a spatially complete snow cover, the IMS Daily Northern Hemisphere Snow and Ice Analysis at 4 km is used to fill in missing data. A pixel is then considered as containing snow if its averaged snow fraction is not zero. In this case, the algorithm considers the pixel is composed by two fractions: the snow-covered fraction and the non-snow-covered fraction, to which channel emissivities are computed (ϵ_{snow} and ϵ_{nosnow}, respectively). ϵ_{BS} and ϵ_{veg} values for land cover type "Snow / Ice" are used to estimate the snow-covered fraction emissivities. The final channel emissivity is given by:

$$\boldsymbol{\epsilon = SCE \times \epsilon_{snow} + (1 - SCE)\epsilon_{nosnow}} \qquad \textbf{(3)}$$

where SCE is the snow cover extent.

Both calibration and verification databases of the GSW developed for each sensor rely on radiative transfer simulations of TOA brightness temperatures for the split-window channels available in the considered sensors. The simulations are performed for the database of global profiles of temperature, moisture, and ozone compiled by Borbas et al. (2005) for clear sky conditions and referred hereafter as SeeBor. The database contains over 15,700 profiles taken from other datasets, such as NOAA88 (Seemann et al., 2003), Thermodynamic Initial Guess Retrieval (TIGR)-like (Chevallier, 2001), and TIGR (Chedin et al., 1985), that are representative of a wide range of atmospheric (clear sky) conditions over the whole globe. In addition, surface parameters such as skin temperatures (TSkin) and a land cover classification within the International Geosphere-Biosphere Programme ecosystem categories (IGBP) (Belward, 1996) are assigned to each profile. Skin temperature over land surfaces corresponds to LST in SeeBor and is estimated as a function of 2-meter temperature (T2m), biosphere information, and solar zenith and azimuth angles (Borbas et al. 2005). In the design of the calibration database, random viewing angles within the range of acceptable usage are assigned to each profile – viewing zenith angle values above 70º are not used in the CGLOPS LST merged product (see section 3.3), as the signature of surface emitted radiation (and therefore LST) weakens for very high viewing angles.

The calibration methodology used here follows the study by Martins et al. (2016). The SeeBor database described above was split into two subsets – one used for the calibration of the GSW algorithms, and an independent one used for verification of the fitted versions. The calibration subset strongly influences the robustness of the model coefficients; therefore, the choice of the profiles requires some caution. The SEVIRI algorithm was calibrated within the framework of LSA-SAF using 77 geographically well-distributed atmospheres, covering a broad variety of water vapour content (from very dry to moist conditions), leaving more than 15,600 profiles for GSW verification (Freitas et al., 2012); this corresponds to the algorithm currently used operationally to generate LSA-SAF LST products. The calibration of ABI and AHI LST algorithms make use of a new and more representative selection of profiles, taken for areas outside the MSG disk, which lead to a lower regression error (Martins et al., 2016). The composition of such calibration database follows the steps below:

i) Define classes of T_{Skin} (from 200 K to 335 K in steps of 5 K) (Figure 1b) and TCWV (from 0 to 6 cm in classes of 0.75 cm (Figure 1a) —values greater than this should be treated with the coefficient corresponding to the last TCWV class).

ii) Iterate in the SeeBor clear sky profile database to fill each class in the TCWV/TSkin phase space (as in Figure 1c) with one case each. When a new profile is selected, it is ensured that its great-circle distance to the already selected profiles is greater than an initial distance of 15 degrees, which guarantees a wide geographical coverage. After a sufficiently large number of tries (in this case 30,000), the distance criterion is relaxed in steps of minus 1 degree, until the whole TCWV/TSkin phase space is filled. The final set has 116 profiles (Figure 1d).

iii) For each of the previously selected profiles, assign a new T_{Skin} based on the ranges of $T_{Skin} - T_{air}$ observed range of this difference. The choice of the range of perturbations to apply is key to the performance of the chosen model and may depend on the region of interest. In the case of this work, a range of ±15 K around T_{air} in steps of 5 K showed an overall good performance.

iv) Each of these conditions may be sensed from angles ranging from 0 (nadir view) to 70° in steps of 2.5°. It is important to discretize the viewing geometry in this way because this is an intrinsically non-linear problem. The upper limit of the viewing zenith angle might be adapted for the sensor under analysis. Previous calibration exercises show that above this viewing angle limit the retrieval errors are generally too high (i.e. above 4 K), especially for moister atmospheres (Freitas et al., 2012).

v) For the emissivity, a range of possible values are attributed to each of the cases above: values of ϵ_{TIR1} from 0.93 to 1.0 in steps of 0.01 and then prescribe departures from this value for ϵ_{TIR2}: -0.015 to 0.035 in steps of 0.01 (excluding cases where $\epsilon_{TIR2} > 1$), as suggested by the distribution of this difference within the SeeBor dataset.

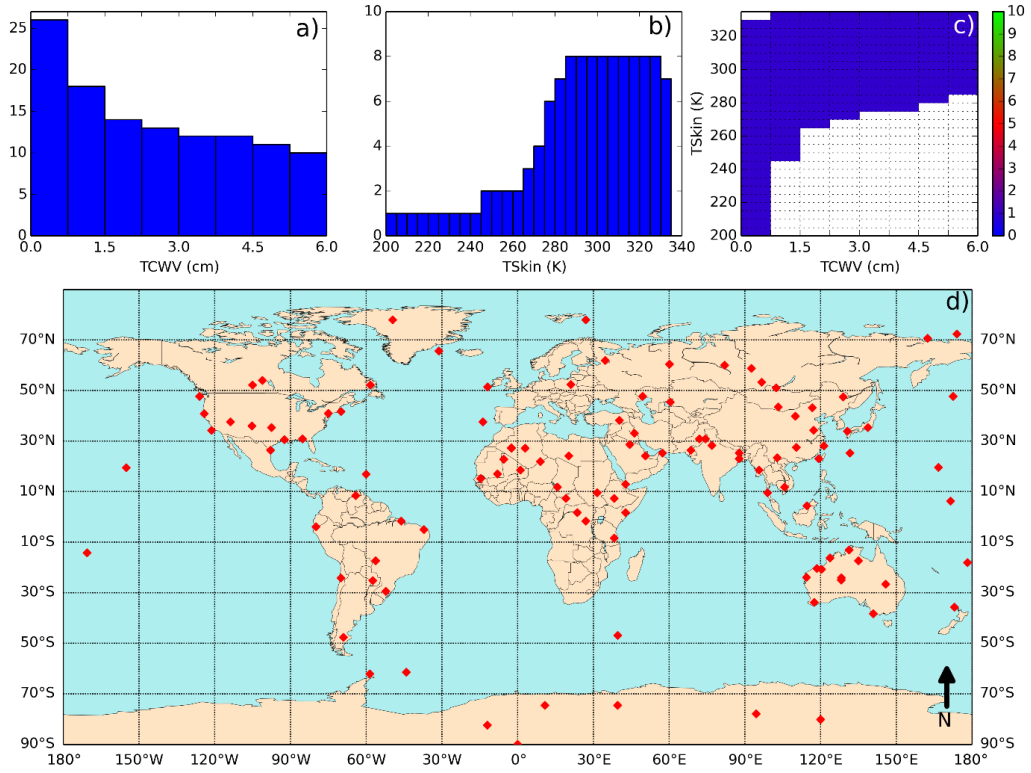


Figure 1: Main properties of the calibration database atmospheric profiles used in the determination of the model coefficients for GOES and Himawari. (a) TCWV distribution; (b) T_{Skin} distribution; (c) bivariate $TCWV/T_{Skin}$ distribution and (d) geographical distribution.

The radiative transfer simulations are performed using the MODerate spectral resolution atmospheric TRANsmittance algorithm (MODTRAN6; Berk et al., 2014). The radiance (L_ν) that would be

measured by each TIR channel is then estimated with MODTRAN6, for each of the surface and atmospheric conditions, and viewing geometry described in the previous paragraph. The simulations are performed using a spectral resolution of 1 cm^{-1}. The channel effective radiance is then derived from the integration of L_ν weighted by the channel response function. The channel radiances are then converted into brightness temperatures using the respective recommend procedure by the satellite agency, i.e. following the GOES Product User Guide [NOAA, 2019] the Himawari Standard Data User's Guide [JMA, 2017].

The GSW parameters A_i, B_i and C obtained for SEVIRI by fitting equation ((1) to the calibration dataset and the variance of LST explained by the regression are available in the respective product ATBD (SAF/LAND/IM/ATBD_MLST/1.2).

In the case of AHI and ABI, ten different calibration/validation databases (of radiative transfer simulations) were computed for each sensor. Each database used a different set of atmospheric profiles obeying the distribution described above (i.e., filling the TCWV/SZA "phase space"), chosen randomly. The adopted set (illustrated in Figure 1, common to both AHI and ABI) was chosen for being the one combining a relatively smooth distribution of model coefficients with good validation statistics in the TCWV/SZA space (since the overall performance is reasonably similar among databases). This distribution is shown in

Figure 2. Since the channel response functions of ABI and AHI are reasonably similar and the set of profiles used for calibration are the same, the coefficients for AHI are also very similar to those of ABI and therefore are not shown here for the sake of brevity.

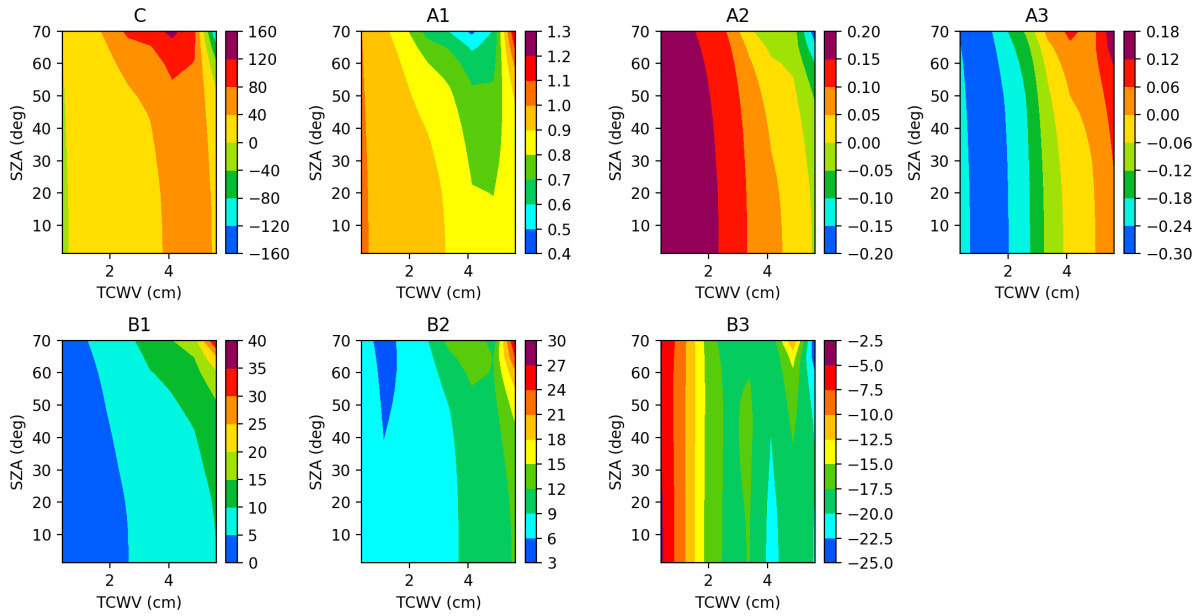


Figure 2: Distribution of the GSW parameters used in ABI/GOES16 (indicated at the top of each panel) as a function of TCWV (in cm) and SZA.

Figure 3 presents the root mean square difference (RMSD) of the GOES-16, MSG/SEVIRI and Himawari GSW as a function of satellite zenith angle and TCWV classes. Differences between the

GSW algorithm developed to the different sensors arise from distinct channel response functions, instrument noise, and differences on the calibration database. Particularly in the case of MSG, we use LSA-SAF LST directly, which used a less comprehensive calibration database. For GOES and Himawari, the calibration database was expanded using the strategy described in Martins *et al*. (2016).

Although some key steps are taken to increase the harmonization of the product, a more harmonized treatment for all sensors is envisaged in future versions. Namely, with the introduction of the Meteosat Third Generation (MTG), all sensors will be calibrated using the same strategy.
The GSW flow chart is shown in Figure 4.

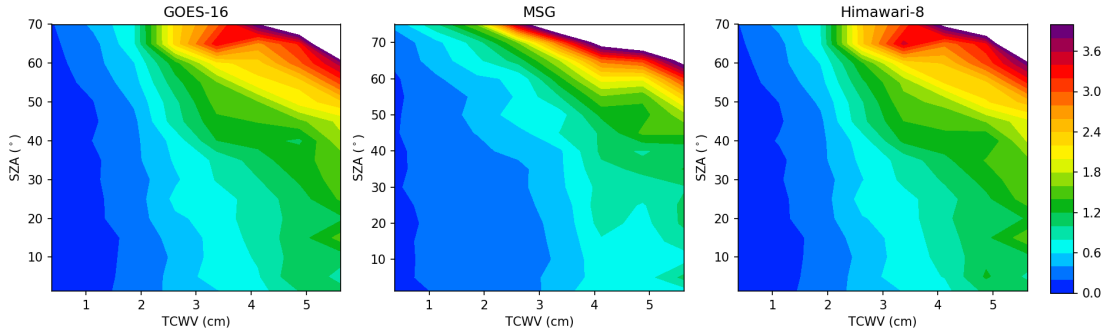


Figure 3: Model RMSD (in K) for the GSW developed for GOES-16 (left), SEVIRI (center) and Himawari-8 (right) thermal infrared window channels.

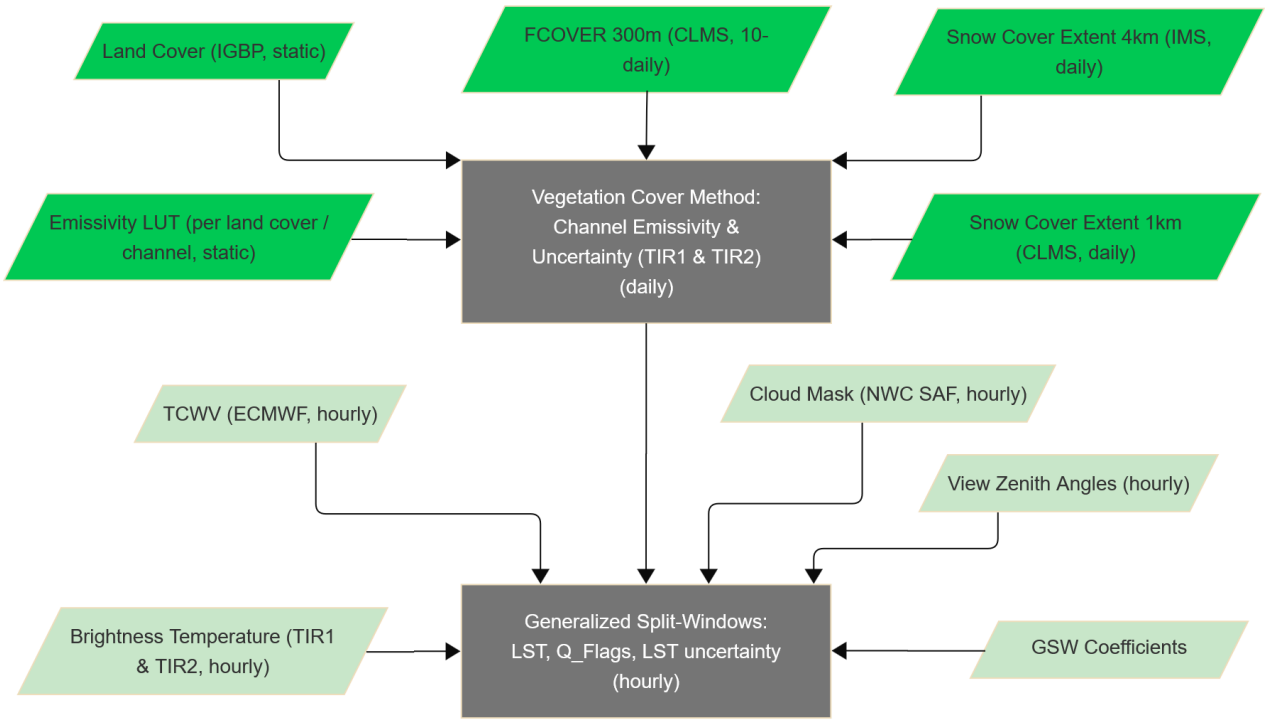


Figure 4: Flow diagram of the generation of LST, its Quality Flags (Q_FLAGS) and associated uncertainty, using the GSW algorithm for GOES and Himawari. Channel emissivity is estimated using

the Vegetation Cover Method. Dark green boxes indicate inputs used for the estimation of channel emissivity, light green boxes indicate inputs to the LST retrieval and grey box indicate algorithm steps.

3.2.3.3 LST retrieval uncertainty

The estimation of the uncertainty for each LST value is based on the framework developed by Freitas et al. (2012) for the case of the LSA-SAF LST product. The method relies on clear identification of the uncertainty sources associated to inputs and to the regression model, and their propagation to the final product, which is detailed below. Surface emissivity, atmospheric total column water vapor, the viewing geometry and, to a lesser extent, the sensor radiometric noise are the main factors affecting the uncertainty in the LST retrieval. Barren soils and sparsely vegetated surfaces are characterized by relatively low emissivities values and high uncertainties, since the variability of emissivity among pixels within these land cover types is not well-captured in the maps used in LST retrievals. Emissivity errors are higher under dry atmospheres, which further contribute to IR retrievals being particularly challenging over arid and semi-arid regions. The presence or absence of snow introduces another source of uncertainty in the retrieval. However, it can be considered as part of the uncertainty due to land cover classification, which is not treated explicitly in the current framework, but is generally captured in the uncertainty due to surface emissivity. Snow and ice have important directional and spectral effects on surface emissivity, whose magnitude depends mainly on snow granularity or viewing geometry. Furthermore, for the “Snow / Ice” land cover class, emissivity values for the TIR2 channel show considerably higher variability when compared to TIR1 (see Table 13 to Table 15 in Annex 2: Emissivities and their uncertainties), which may also affect the retrieval quality. This type of uncertainty could be minimized if there was a robust way to distinguish snow and ice surfaces (Snyder *et al.*, 1998). Another important aspect to consider is the uncertainty due to the time mismatch between the last snow cover extent observation and the LST retrieval: the CLMS SCE is a daily product and therefore provides the snow cover from the previous day. This means that snow might have fallen over a given area while the snow mask describes the pixel as snow-free. Or that snow might have melted while the snow mask still classifies the surface as snow-covered. It is worth mentioning that snow occurs on periods of high cloudiness, so LST retrievals are rarely possible on such occasions, and therefore the impacts of the abovementioned situations are likely to be small.

Furthermore, high atmospheric moisture content enhances the non-linear effects in atmospheric absorption/emission of the IR radiation and increases the optical path, limiting the retrieval by a linear regression scheme such as the GSW (in these cases, the model regression uncertainty is also high – see Figure 3). The optical path is also increased for high satellite viewing angles, and therefore the uncertainty generally increases towards the satellite disk edge.

Finally, errors in the cloud masking algorithm may severely affect the LST retrieval and are usually characterized as negative outliers when compared to in-situ values in mostly any product validation exercise. According to validation results of the EUMETSAT SAF on Nowcasting and Very Short-Range Forecasting (NWC SAF) cloud mask for SEVIRI, the expected rate of missed clouds is of the order of 4% [NWC-CDOP2-GEO-MFL-SCI-VR-Cloud_v1.0]. These missed cases often correspond

to broken clouds or cases in neighbouring cloudy pixels. It is very difficult to propagate the uncertainty in cloud identification to LST error bars (Bulgin *et al.*, 2018).

In a real scenario, we do not have access to the exact GSW inputs $X = (T_{10.5}, T_{12.3}, \epsilon_{10.5}, \epsilon_{12.3})$ and $Y = (W, \Psi)$ (where W denotes the TCWV and Ψ is the viewing zenith angle), but only to inaccurate inputs, which we denote by $\hat{X} = (\hat{T}_{10.5}, \hat{T}_{12.3}, \hat{\epsilon}_{10.5}, \hat{\epsilon}_{12.3})$ and $\hat{Y} = (\widehat{W}, \widehat{\Psi})$. Therefore, if we still infer the LST according to model (1) replacing the exact GSW inputs with the inaccurate ones, we have a new source of uncertainty on the top of the fitting error ΔLST. In the current section, the main uncertainty sources are identified and their impact on the total LST uncertainty is estimated.

Let us define the vector of model coefficients $\theta = (A_1, A_2, A_3, B_1, B_2, B_3, C)$. Notice that the vector θ generated by the fitting process is a function of TCWV and viewing zenith angle i.e., (Y) . Consider the LST estimator $\widehat{LST} = f(\hat{X}, \hat{\theta})$, where $\hat{\theta} = \theta(\hat{Y})$ and $f(X, \theta)$ is the LST estimate given by model (1). A characterization of the model uncertainty is given by:

$$\boldsymbol{S_{LST} = E\left[\left(f(\hat{X}, \hat{\theta}) - LST\right)^2 \middle| X, Y\right]^{1/2}} \quad \textbf{(4)}$$

where the operator $E[.|X, Y]$ stands for mean value conditioned to X and Y; i.e. for a given GSW input X, Y, we want to compute the RMSD of the LST estimate. Using the fact that $LST = f(X, \theta) + \Delta LST$ and assuming that $E\left[f(\hat{X}, \hat{\theta})|X, Y\right] = f(X, \theta)$, we may write:

$$\boldsymbol{S_{LST}^2 = E\left[\left(f(\hat{X}, \hat{\theta}) - f(X, \theta)\right)^2 \middle| X, Y\right]^{1/2} + \Delta LST^2} \quad \textbf{(5)}$$

By taking a linear approximation of $f(\hat{X}, \hat{\theta})$ in the neighbourhood of (X, θ) and denoting $\sigma_{X_i}^2 = E\left[\left(\hat{X}_i - X_i\right)^2 | X\right]$ and $\sigma_{\theta_i}^2 = E\left[\left(\hat{\theta}_i - \theta_i\right)^2 | Y\right]$, we are led to:

$$\boldsymbol{S_{LST}^2 = \sum_i \left(\frac{\partial f}{\partial X_i}\right)^2 \sigma_{X_i}^2 + \sum_j \left(\frac{\partial f}{\partial \theta_j}\right)^2 \sigma_{\theta_i}^2 + \Delta LST^2} \quad \textbf{(6)}$$

where we have assumed that the components of X, Y are mutually independent and that the $E\left[\left(\hat{X}_i - X_i\right)| X\right] = 0$ and $E\left[\left(\hat{\theta}_i - \theta_i\right)| Y\right] = 0$. Next, we study in detail the error due to each individual GSW input.

3.2.3.3.1 Impact of sensor noise on LST uncertainty

The impact of sensor noise on LST is given by the sum of the uncertainties associated to the noise of each split-window channel:

$$S_T^2 = S_{T_{IR1}}^2 + S_{T_{IR2}}^2 \quad (7)$$

where:

$$S_{T_{IR1}}^2 = \left(\frac{\partial f}{\partial T_{IR1}}\right)^2 \sigma_{T_{IR1}}^2 \text{ and } S_{T_{IR2}}^2 = \left(\frac{\partial f}{\partial T_{IR2}}\right)^2 \sigma_{T_{IR2}}^2 \quad (8)$$

In this case, the value of $\sqrt{\sigma_{\mathrm{T_{IRn}}}^2}$ is given by the instrument radiometric noise, $NE\Delta T$, to be reported for each channel upon instrument calibration. In the case of the currently operational sensors in use for the CGLOPS LST, all channels / instruments have a reported $NE\Delta T = 0.1$K.

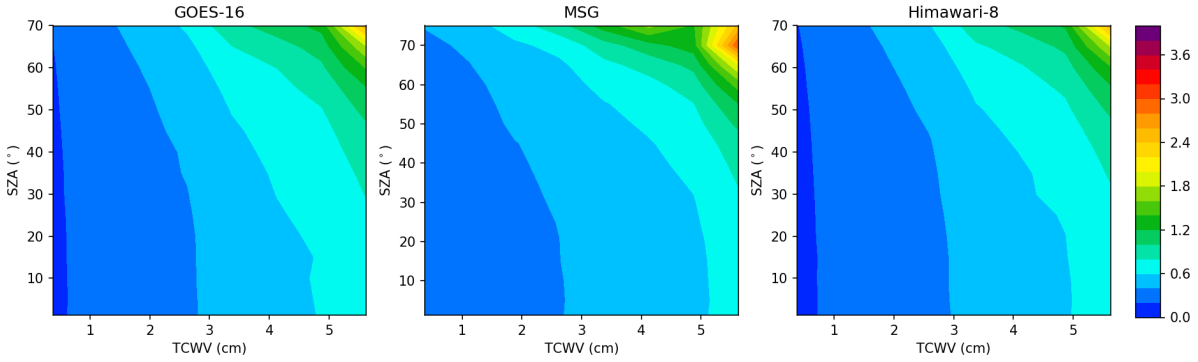
In the case of the GSW algorithm [Eq.(1)], the derivatives with respect to the split-window brightness temperatures are given by:

$$\frac{\partial f}{\partial T_{IR1}} = \frac{1}{2}\left[A_1 + B_1 + (A_2 + B_2)\frac{1-\epsilon}{\epsilon} + (A_3 + B_3)\frac{\Delta\epsilon}{\epsilon^2}\right] \quad (9)$$

$$\frac{\partial f}{\partial T_{IR2}} = \frac{1}{2}\left[A_1 - B_1 + (A_2 - B_2)\frac{1-\epsilon}{\epsilon} + (A_3 - B_3)\frac{\Delta\epsilon}{\epsilon^2}\right] \quad (10)$$

The expected impact of this uncertainty is assessed here by perturbing each of the cases in the validation database with 200 random normally distributed perturbations to each T_{IR_i} and then computing the variance of the differences between the perturbed vs. unperturbed inputs. The result is shown for each sensor, per class of TCWV and SZA, in

Figure 5. GOES and Himawari show similar behaviour, with impacts growing towards higher optical paths, while MSG shows a slightly higher impact towards higher optical paths, compared to GOES and Himawari.



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Figure 5: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to sensor noise, S_T, as a function of TCWV and SZA.

3.2.3.3.2 Impact of surface emissivity uncertainties

The impact of uncertainties in surface emissivity for both split-window channels is given by:

$$S_{\epsilon}^2 = S_{\epsilon_{IR1}}^2 + S_{\epsilon_{IR2}}^2 \quad (11)$$

where

$$S_{\epsilon_{IR1}}^2 = \left(\frac{\partial f}{\partial \epsilon_{IR1}}\right)^2 \sigma_{\epsilon_{IR1}}^2 \text{ and } S_{\epsilon_{IR2}}^2 = \left(\frac{\partial f}{\partial \epsilon_{IR2}}\right)^2 \sigma_{\epsilon_{IR2}}^2 \quad (12)$$

where $\sigma_{\epsilon_{IR1}}$ and $\sigma_{\epsilon_{IR2}}$ are the uncertainty of the respective channel emissivities. To estimate $\sigma_{\epsilon_i}^2$, we need to take into account the uncertainties due to ϵ_{veg_i}, ϵ_{bs_i} and FVC used in Eq. ((2). Thus, by propagating their uncertainties, we get:

$$\sigma_{\epsilon_{IR_i}}^2 = FVC^2 \sigma_{\epsilon_{veg_i}}^2 + (1 - FVC)^2 \sigma_{\epsilon_{bs_i}}^2 + \left(\epsilon_{veg_i} - \epsilon_{bs_i}\right)^2 \sigma_{FVC}^2 + \left(d\epsilon_{MR_{IR_i}}\right)^2 \quad (13)$$

where $d\epsilon_{MR_{IR_i}}^2$ is the uncertainty due to multiple reflections. This factor is added here as a source of uncertainty due to the difficulty to explicitly model this phenomenon, which would depend on geometrical factors such as the canopy height and typical separation between vegetation elements (Trigo *et al.*, 2008a). Its effect on the emissivity uncertainty for each channel is estimated using:

$$d\epsilon_{MR_{IR_i}} = 4\langle d\epsilon \rangle_{IR_i} FVC(1 - FVC) \quad (14)$$

Tabulated values of $\langle d\epsilon \rangle_{IR_i}$ per IGBP land cover type (Table 12 of Annex 2: Emissivities and their uncertainties.) and per channel were calculated in Peres and DaCamara (2005) and are reproduced in Table 2.

Table 2: Values used for $\langle d\epsilon \rangle$ for each split-window channel (see equation (14)). For land covers that are not in the table, values are zero.

IGBP Class	TIR1	TIR2
1; 2	0.0139	0.0123
3; 4	0.0138	0.0122
5	0.0139	0.0122
6	0.0102	0.0088
7	0.0022	0.0019
8	0.0101	0.0087
9	0.0060	0.0052
10	0.0012	0.0011
12	0.0059	0.0048
13	0.0016	0.0011
14	0.0070	0.0060
16	0.0020	0.0013

In the presence of snow, that source of uncertainty needs to be considered. This extra term is computed for pixels with snow/ice land cover and for pixels where some snow is detected (i.e., when SCE > 0.). The CLMS SCE Quality Assessment Report (QAR) [CGLOPS2_QAR_SCE-NHEMI-1km_V1] reports a mean unbiased RMSD of 13 %, therefore this value is assumed for σ_{snow}. Assuming zero uncertainty due to SZA, and based in Eq. (3), the uncertainty due to snow is estimated as:

$$\boldsymbol{\sigma^2_{\epsilon_{IR_i}} = SCE^2 \sigma^2_{\epsilon_{snow_i}} + (1 - SCE)^2 \sigma^2_{\epsilon_{nosnow_i}} + \left(\epsilon_{snow_i} - \epsilon_{nosnow_i}\right)^2 \sigma^2_{snow},} \quad \textbf{(15)}$$

where $\sigma^2_{\epsilon_{snow_i}}$ is estimated considering the variability of snow/ice spectra for the materials in the spectral database (see values in Table 13 to Table 15 for land cover type 15), and $\boldsymbol{\sigma^2_{\epsilon_{nosnow_i}}}$ is $\boldsymbol{\sigma^2_{\epsilon_{IR_i}}}$ from Eq. (13). Finally, the derivatives in Eq. (12) are given by:

$$\frac{\partial f}{\partial \epsilon_{IR1}} = \frac{1}{2}\left[-\frac{A_2}{2}\frac{T_{IR1}+T_{IR2}}{\epsilon^2} + A_3\,(T_{IR1}+T_{IR2})\left(\frac{\epsilon-\Delta\epsilon}{\epsilon^3}\right) - B_2\,\frac{T_{IR1}-T_{IR2}}{\epsilon^2} + B_3\,(T_{IR1}-T_{IR2})\left(\frac{\epsilon-\Delta\epsilon}{\epsilon^3}\right)\right] \quad \textbf{(16)}$$

$$\frac{\partial f}{\partial \epsilon_{IR2}} = -\frac{1}{2}\left[\frac{A_2}{2}\frac{T_{IR1}+T_{IR2}}{\epsilon^2} + A_3\,(T_{IR1}+T_{IR2})\left(\frac{\epsilon+\Delta\epsilon}{\epsilon^3}\right) + B_2\,\frac{T_{IR1}-T_{IR2}}{\epsilon^2} + B_3\,(T_{IR1}-T_{IR2})\left(\frac{\epsilon+\Delta\epsilon}{\epsilon^3}\right)\right] \quad \textbf{(17)}$$

In Figure 6, an assessment of this source of uncertainty is shown as a function of TCWV and for 3 classes of mean emissivity (between TIR1 and TIR2): arid ($\epsilon \leq 0.96$), sparsely vegetated ($0.96 < \epsilon \leq 0.98$) and highly vegetated/water ($\epsilon > 0.98$). For each profile of the validation database, a reference retrieval was made using the FVC, ϵ_{bs} and ϵ_{veg} values of Table 13 to Table 15, corresponding to the land cover assigned to the profile. Then, 200 random normally distributed perturbations of FVC were introduced in each case of the validation database with a standard deviation equal to the reported uncertainty in FVC, as well as 200 uniformly distributed perturbations of ϵ_{bs} and ϵ_{veg} with distribution half-widths corresponding to the respective uncertainty. Results for GOES and Himawari are very similar, with 1) a general decrease of S_ϵ with TCWV due to compensation of errors between emitted land surface radiance and reflected radiance emitted by the atmosphere (which increases in moister atmospheres), and 2) with larger values in situations with less vegetation, which are characterized by lower and more uncertain emissivities. MSG shows somewhat higher values (except for emissivity values characteristic of sparsely vegetated surfaces), not only due to differences in the calibration database but also due to differences in the channel frequencies/response functions of each channel. Bare ground emissivity uncertainties are higher for sparsely vegetated land covers (e.g., savannas) but are relatively lower at the 10.8 µm channel used by MSG when compared to the 11.2 µm channel used by both ABI and AHI). The same is true for the TIR2 channels (see tables in Annex 2: Emissivities and their uncertainties).

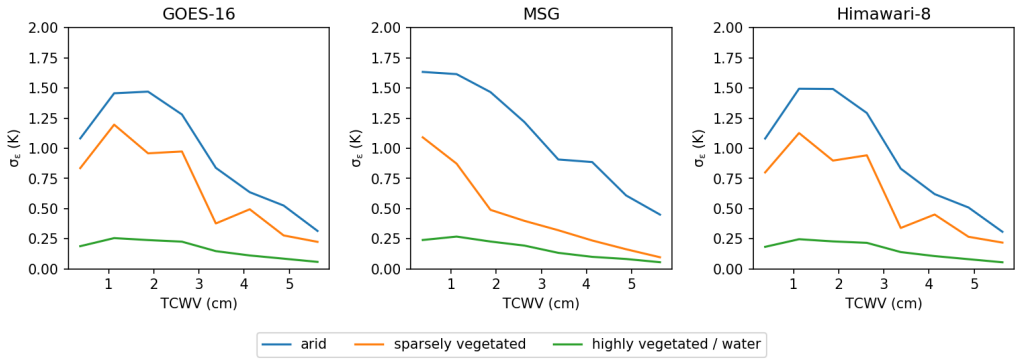


Figure 6: Variability of LST for GOES-16 (left), MSG (center) and Himawari-8 (right) due to emissivity uncertainty, S_ϵ^2, as a function of TCWV. Values are given for 3 classes of mean emissivity (between TIR1 and TIR2): arid ($\epsilon \leq 0.96$) in blue, sparsely vegetated ($0.96 < \epsilon \leq 0.98$) in yellow and highly vegetated/water ($\epsilon > 0.98$) in green.

3.2.3.3.3 Uncertainties in forecasts of atmospheric water vapour content

Since the total column water vapor is used implicitly in the algorithm (i.e., different sets of parameters are used for each class of TCWV values), the uncertainty cannot be calculated analytically. The uncertainty due to this parameter is related to the fact that there is a probability of choosing the wrong

set of model coefficients because the TCWV forecast may lead to a wrong choice of TCWV class. Therefore, it is estimated *a priori* as follows:

1) the operational use of the GSW algorithm (Eq.(1)) to retrieve LST from each sensor makes use of forecasts of TCWV (W) provided by the ECMWF. To characterize W error statistics, we compare ECMWF W forecasts (with forecast steps ranging between 12 and 36 h) with the respective analysis, for the 15th of each month for one full year; ECMWF grid points with model cloud cover higher than 10% were excluded. This exercise is repeated regularly (about once per year) to update the uncertainty in ECMWF forecasts. The comparison between W forecasts and analysis (the reference value) allows us to estimate the probability $P\left(\widehat{W}_{i_{fc}}|W_{i_{an}}\right)$, i.e., the probability that $\widehat{W}$ belongs to the water vapour content class $W_{i_{fc}}$, given that the true class is $W_{i_{an}}$.
2) The validation database is used to calculate the variance of the LST for each class of SZA/TCWV. The variance is obtained by simulating each case of the validation database (i_{valid}) with all the possible sets of coefficients corresponding to that SZA (i_{fc}) and then comparing to the value obtained with the "true" set of coefficients ($LST_{i_{valid}}$). The uncertainty final estimate is provided for each SZA / forecast TCWV class by multiplying the obtained variance by the probability of a given forecast class does not correspond to the "true" class:

$$\boldsymbol{S_W^2 = \frac{1}{n_{TCWV}} \sum_{i}^{n_{TCWV}} \sigma^2_{\widehat{W}_{i_{an}}|W_{fc}} P(W_{i_{an}}|\widehat{W}_{fc})} \quad \textbf{(18)}$$

where $\sigma^2_{\widehat{W}_{i_{an}}|W_{fc}}$ is the variance of the errors of LST for a given W estimate (i.e., for a given TCWV forecast $\widehat{W}_{fc}$), when the correct value is $W_{i_{an}}$; $P\left(W_{i_{an}}\middle|\widehat{W}_{fc}\right)$ is the probability that the correct class of TCWV is determined by $W_{i_{an}}$, conditioned by the forecast value of $\widehat{W}_{fc}$; n_{TCWV} is the number of TCWV classes considered.

In Figure 7, the variability due to uncertainty in LST is shown as function of TCWV and SZA. Again, higher values occur towards higher optical paths (i.e., towards higher TCWV and higher SZA). In general, the uncertainty due to TCWV is small because the forecasts are generally close to the analysis used for verification (therefore it is unlikely that a set of coefficients corresponding to a different TCWV class is used, especially towards lower TCWV values). Only for those classes with higher TCWV, the fact that a different set of coefficients is used will imply higher differences in the retrieved LST.

These values are prescribed to the operational chain as a LUT, since it is not possible to perform these calculations for each forecast due to the unavailability of the analysis at the time of production.

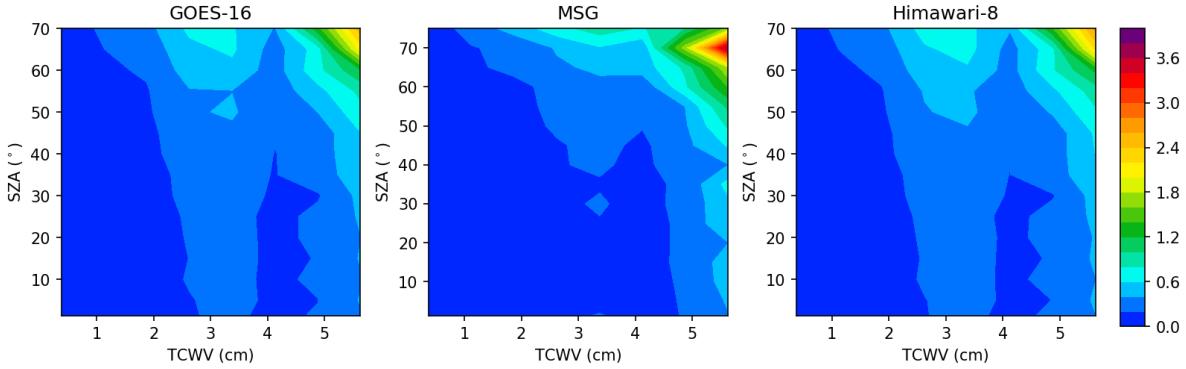


Figure 7: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to the uncertainty in TCWV forecasts, S_W, as a function of TCWV and SZA.

3.2.3.3.4 Total uncertainty of LST retrievals

The estimation of LST error bars, S_{LST}, assumes that all sources of errors described in the previous sections are independent and therefore is given by:

$$S_{LST} = \sqrt{S_T^2 + S_\epsilon^2 + S_W^2 + \Delta LST^2} \quad (19)$$

In Figure 8, an assessment of the total expected uncertainty is shown as a function of TCWV and SZA, for each sensor and for arid, sparsely vegetated and highly vegetated conditions. Overall, uncertainty grows towards higher optical paths, to the point that for a few classes of high TCWV and SZA the usage of the product is not recommended and therefore all situations with pixel uncertainty higher than 5 K are not considered for the final LST product (see section 3.3). The required product target uncertainty is 2 K (see Chapter 2). However, users are advised to take into consideration the estimated error bars for their application. LST values with uncertainties above 4 K must be used with caution. These values are usually found in those classes with higher optical path (see white areas in Figure 8).

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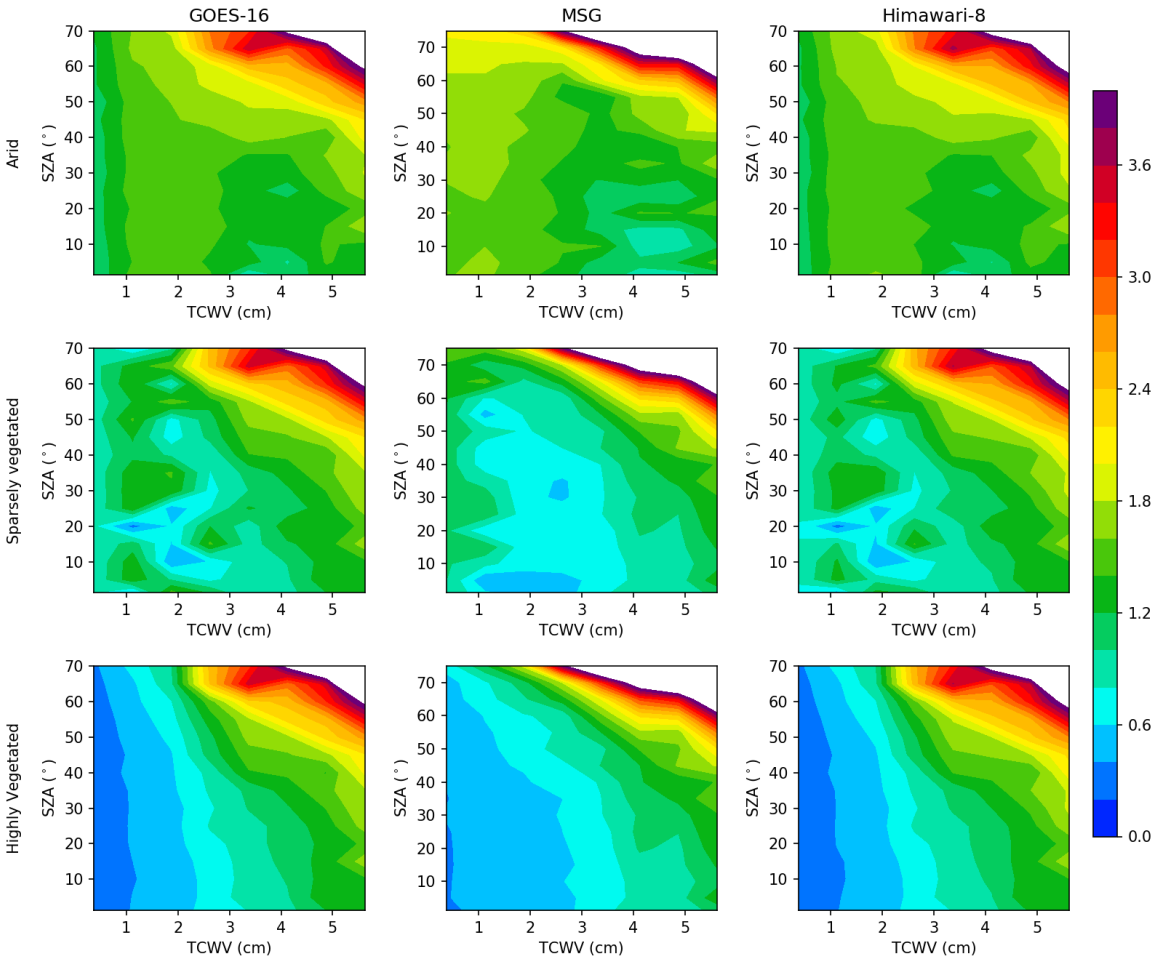


Figure 8: Variability of LST (in K) for GOES-16 (left), MSG (center) and Himawari-8 (right) due to the uncertainty in all inputs, S_{LST}, as a function of TCWV and SZA. From top to bottom, results are shown for arid, sparsely vegetated and highly vegetated situations. White areas correspond to uncertainties above 4 K.

3.3 DATA FUSION

LST corresponds to instantaneous fields produced every 15 minutes in the case of SEVIRI, and hourly in the case of GOES and Himawari. The three individual fields are then re-projected and merged onto a 0.025º regular grid. The individual timeslots selected to integrate the global product are T=0 for all sensors. However, discrepancies may arise from the different configurations of the scan: SEVIRI's full-disk scan takes about 12 minutes and is performed from south to north, while ABI/AHI's scan can take between 5 to 10 minutes, depending on the mode, and is performed from north to south.

LST data fusion consists of LST retrievals from the different GEO satellites merged into a single field, thus yielding a global product (see Figure 9). The final LST product is obtained by interpolating each GEO LST product to the target global grid (regular in latitude and longitude) with a nearest

neighbour method, for SZAs up to 70º for all sensors. In the case of overlap between sensors (e.g South America between GOES and MSG), a weighted average considering each sensor viewing angle is computed so that the final product is less influenced by lower quality retrievals (i.e., those with greater SZA).

Additional information crucial for the product applications are included in the LST product: i) the error bars in a pixel by pixel basis, determined considering algorithm uncertainties, as well as the propagation of input uncertainties (section 3.2.3.3; Freitas et al., 2012); ii) the pixel acquisition time – in areas observed by more than one satellite, this value corresponds to an average of acquisition times, also weighted by the SZA; and iii) a quality flag with satellite information, missing data, cloudiness and land-water mask (Table 16 in Annex 3: LST Quality Control Information).

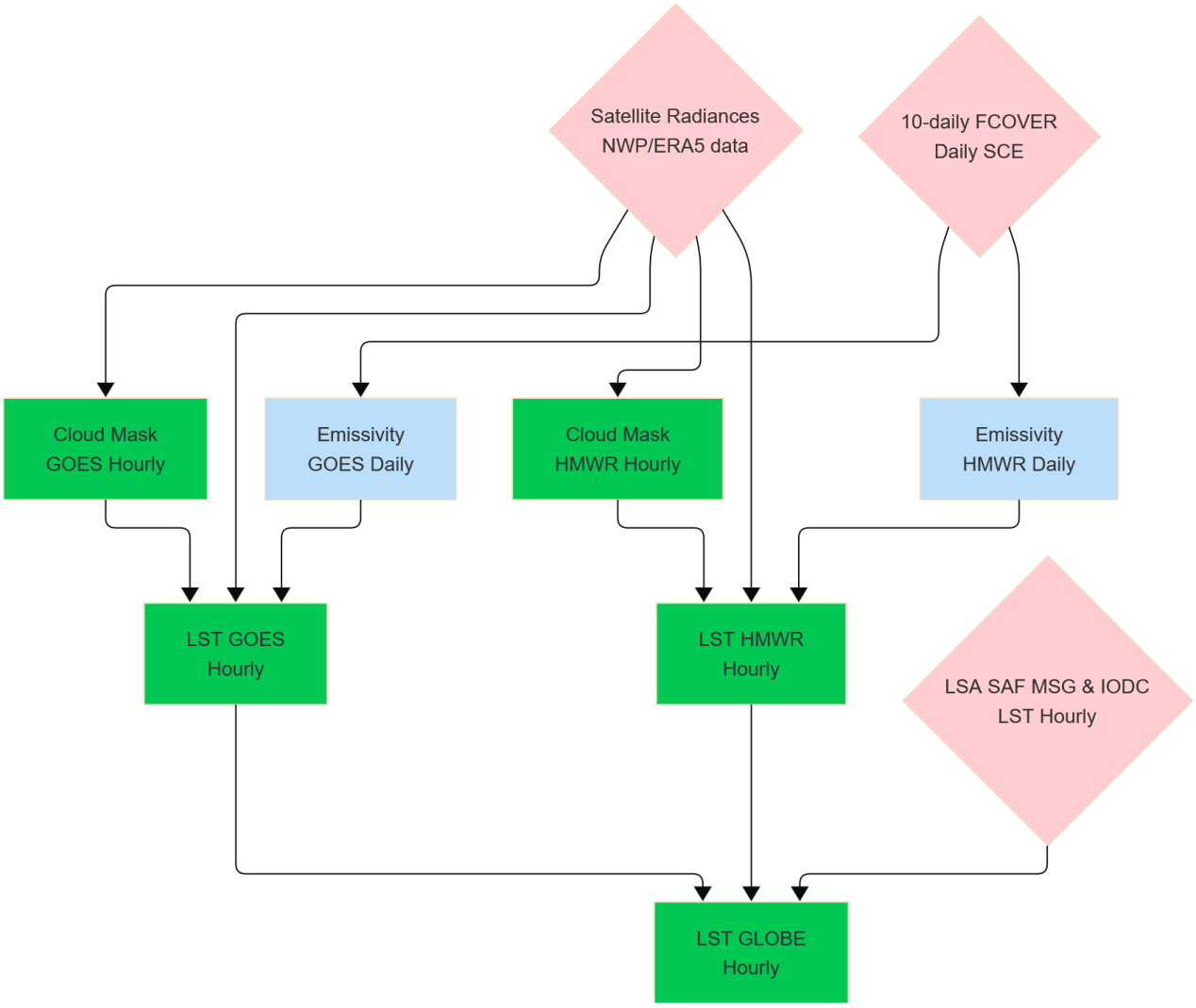


Figure 9: Flow diagram of the generation of LST GLOBE product.

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4 KNOWN LIMITATIONS

The vegetation cover method has been widely used to estimate surface emissivity in the TIR bands used for LST retrievals requiring this information (Becker and Choudhury, 1988; Caselles *et al.*, 1997; Peres and DaCamara, 2005; Trigo *et al.*, 2008a). This method assumes that a given bare soil emissivity, ϵ_{bg}, and a given vegetation emissivity, ϵ_{veg} for each land cover class are representative of the whole land cover class. This is not always the case, e.g., in croplands, urban or barren areas, where differences in soil composition, crop/vegetation density or presence of man-made infrastructures may indeed correspond to different surface emissivities within the same land cover class. In the algorithm used here, this emissivity variability within each land cover class is accounted for by prescribing adequate uncertainties for bare soil and vegetation emissivities (σ_{bs} and σ_{veg}, respectively). These values are obtained by estimating the maximum absolute deviation of the emissivities (both ϵ_{bs} and ϵ_{veg}) within each class, using all the materials from the spectral emissivity database assigned to that class. Consequently, this method provides an upper estimate of the uncertainty due to emissivity variability within each land cover class. The more comprehensive the spectral database is, the more accurate is the uncertainty estimate using this method.

Careful analysis of Table 13, Table 14 and Table 15 shows that the relative uncertainty of emissivities within each class is below 1 %, except for land cover types 15 and 16 (Snow/Ice and Barren), where it may reach 3 %. The effect of vegetation cover (i.e., due to the differences between ϵ_{bs} and ϵ_{veg} within each class) for all classes is of the order of 3%. Accounting for the effect of vegetation cover was key to reduce the uncertainty due to emissivity in all vegetated land cover classes. The methods to improve the estimation of bare soil emissivity are subject of current research (e.g., Borbas *et al.*, 2018; Hulley *et al.*, 2015; Jiménez-Munoz *et al.*, 2014; Malakar and Hulley, 2016; Masiello *et al.*, 2018) and may be considered for future versions of the CLMS LST product after they reach an acceptable maturity level.

5 OUTPUT PRODUCT

The LST files contain one main dataset corresponding to LST estimations, a dataset with the associated LST uncertainties, a layer with the data acquisition time, and a quality flag dataset. The latter consists of a 16-bit unsigned integer assigned as described in Table 16 of Annex 3: LST Quality Control Information.

All the CLMS LST product datasets are generated in a common projection and resolution of 0.025 º which is higher than the 5km requirement [AD1 and Chapter 2]. This resolution increase from the previous version 2 of 0.0.5 is justified since Himawari and GOES have 2 km resolution and MSG maintains 3 km at nadir. This represents a good compromise, offering added value by providing the product at a resolution closer to that of the native satellite imagery. Naturally, the effective resolution decreases with increasing satellite viewing angle, while the regular lat/lon grid results in higher resolution toward the poles. This will be documented in the PUM, including the “effective” resolution for each pixel.

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6 RISK OF FAILURE AND MITIGATION MEASURES

The risks associated with CLMS LST product are here classified with respect to effect and frequency following the standards of *The Open Group*[3].

The effects are assessed using the following criteria:

Catastrophic: infers a complete loss of production and no reprocessing capabilities due to missing mandatory input data.

Critical: infers loss of production in a significant part of the globe (e.g., one satellite disk) due to missing mandatory input data with implication on the service performance but no impact on product quality.

Marginal: the product is generated in a degraded mode with possible implications on the product quality but no implications on the service performance.

Negligible: nominal production over the majority of the potential processed areas or a slightly decrease on the product quality.

The frequency could be enumerated as follows:

Frequent: Likely to occur very often.

Likely: Occurs several times over the course of the CLMS service.

Occasional: Occurs sporadically.

Seldom: Remotely possible and would probably occur not more than once over the course of the CLMS service.

Unlikely: Will probably not occur during the CGLOPS service.

The scheme to assess impact on the LST is defined as:

Extremely High Risk (*E*): The process of global LST will most likely fail with severe consequences.

High Risk (*H*): Significant failure of parts of the LST resulting in certain user requests not being achieved.

Moderate Risk (*M*): Noticeable failure of LST production threatening the success of certain user requests.

Low Risk (*L*): Certain user requests might not be fully successful.

[3] http://www.opengroup.org/

These impacts can be derived using the classification scheme in Table 10.

Table 3: Risk Classification Schema.

Effect	Frequency				
	Frequent	**Likely**	**Occasional**	**Seldom**	**Unlikely**
Catastrophic	*E*	*E*	*H*	*H*	*M*
Critical	*E*	*H*	*H*	*M*	*L*
Marginal	*H*	*M*	*M*	*L*	*L*
Negligible	*M*	*L*	*L*	*L*	*L*

The main risks associated with the LST production are the failures of Hardware/Software (HW/SW) infrastructure and the missing input data. As described in previous sections (3.2.3.1) the global LST product requires the following foreign inputs:

- Global Landcover
- LSA-SAF LST (MSG 0° and IODC)
- GEO imagery
- Forecasted atmospheric data
- CLMS FCOVER 300m product
- CLMS Snow cover extent 1 km product

The risk identification and classification, as well as the mitigation measures are presented in Table 11.

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Table 4: Risk identification and mitigation assessment.

Risk Context	Risk Identification	Effect	Frequency	Impact	Mitigation Measure
HW/SW infrastructure	HW Failure	Catastrophic	Unlikely	***M***	A mirror system ready to start over the nominal service.
	SW errors	Marginal	Occasional	***M***	Operators and developers ready to identify and correct bugs.
Global Landcover	No	-	-	**-**	Since this is a static database there is no risk of failure thus any mitigation measure needed.
LSA-SAF LST	Missing data	Critical	Occasional	***H***	Nominal LST would be produced but only in the regions not exclusively covered by MSG satellites.
	Missing Service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by MSG satellites. The process of LST for the MSG satellites in the framework of CGLOPS could be assessed.
GEO Imagery	Missing GOES16/19 Imagery	Negligible	Occasional	***L***	Use data from backup satellite to produce nominal LST product. LST algorithm prepared to operate with the backup satellite data using a less accurate algorithm.
	Missing GOES service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by GOES satellite.
	Missing Himawari-8 Imagery	Negligible	Occasional	***L***	Use data from backup satellite (Himawari-9) to produce nominal LST product. LST algorithm prepared to operate with the backup satellite data.
	Missing Himawari service	Critical	Unlikely	***L***	Nominal LST would be produced but only in the regions not exclusively covered by Himawari satellite. The use of other satellites e.g. FYI, could be assessed to cover Asia region.
Forecasted atmospheric data	Missing the most recent forecasted data from ECMWF	Negligible	Occasional	***L***	Use the forecasted data from the previous model run with higher step to produce nominal LST.
	Missing ECMWF forecasted data	Marginal	Seldom	***L***	Use ECMWF ERA 5 data to produce LST in degraded mode due to coarser spatial and temporal resolution.
	Missing ECMWF service	Marginal	Unlikely	***L***	Use forecasted data from GFS (NOAA) model to produce LST in degraded mode.
Fraction of Vegetation Cover	Missing the most recent FCOVER RT0	Marginal	Seldom	***L***	Use a RT6-based (6 dekads delay) FCOVER climatology to account for seasonal vegetation variability
Snow cover extent	Missing the most recent Snow Cover Product	Marginal	Seldom	***L***	Filled with IMS product

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ANNEX 1: HISTORY OF CHANGES IN INPUT DATA

This Annex lists the changes in inputs data used in the reprocessing and in the near-real time processing.

Table 5: Input LST of MSG 0° used for the LST GLOBE product

Start to end Date	Version
2018-01-01 to 2023-03-22	Meteosat-11 (MSG4)
2023-03-22 to NRT	Meteosat-10 (MSG3)

Table 6: Input LST of MSG-IODC used for the LST GLOBE product

Start to end Date	Version
2018-01-01 to 2022-06-30	Meteosat-8 (MSG1) 41.5°E
2022-07-1 to NRT	Meteosat-9 (MSG2) 45.5°E

Table 7: Input GOES satellite Brightness temperatures used for the GOES LST retrieval.

Start to end Date	Version
2018-01-01 to 2025-04-07	GOES-16
2025-04-08 to NRT	GOES-19

Table 8: Input Himawari satellite Brightness temperatures used for the Himawari LST retrieval.

Start to end Date	Version
2018-01-01 to 2022-12-12	Himawari 8
2022-12-13 to 2025-10-11	Himawari 9
2025-10-12 to 2025-11-25	Himawari 8 (fall back due to failure in Himawari 9)
2025-11-26 to NRT	Himawari 9 (Himawari 9 resumed operations)

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Table 9: Input vegetation cover used for the derivation of the channel emissivities for GOES and Himawari mission

Start to end Date	Version
2018-01-01 to 2020-06-30	CLMS FCOVER 300m version 1.0 RT6 [IMAGINES_RP2.1_ATBD-LAI300m]
2020-07-01 to 2024-12-31	CLMS FCOVER 300m version 1.1 RT6 [CGLOPS1_ATBD_FCOVER300m-V1.1]
2025-01-01 to 2026-01-28	CLMS FCOVER 300m version 1.1 RT0 [CGLOPS1_ATBD_FCOVER300m-V1.1]
2026-01-29 to NRT	CLMS FCOVER 300m version 2 RT0 [CGLOPS1_ATBD_FCOVER300m-V2]

Table 10: Input snow cover extent used for the derivation of the channel emissivities for GOES and Himawari mission

Start to end Date	Version
2018-01-01- to 2026-02-10	CLMS SCE-NHEMI-1km v1 [CGLOPS2_ATBD_SCE-NHEMI-1km_V1]
2026-02-11 to NRT	CLMS SCE-GLOBE-1km v1 [CGLOPS2_ATBD_SCE-GLOBE-1km_V1]
2018-01-01 to NRT	IMS Daily Northern Hemisphere snow and Ice analysis 4km (GO2156, Version 1)

Table 11: Input total column water vapour used for the GOES and Himawari LST retrieval.

Start to end Date	Version
2018-01-01 to 2025-12-31	ECMWF C3S ERA5
2026-01-01 to NRT	ECMWF Operational Numerical Weather Prediction forecasts 00/12 UTC +12h to +24h

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ANNEX 2: EMISSIVITIES AND THEIR UNCERTAINTIES

Table 12: Land cover type classification used in the product, based on IGBP classification (Belward, 1996).

IGBP Land Cover type	Description
1, 2	Evergreen Forest
3, 4	Deciduous Forest
5	Mixed Forest
6	Closed Shrubland
7	Open Shrubland
8	Woody Savanna
9	Savanna
10	Grasslands
11	Permanent Wetland
12	Cropland
13	Urban and Built-up
14	Cropland Mosaics
15	Snow/Ice
16	Bare soil and rocks
17	Ocean
18	Water bodies

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Table 13: Emissivities and their uncertainties for the GOES-16/19 split-window channels, per land cover type.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9968 ± 0.0018	0.9697 ± 0.0143	0.9975 ± 0.0025	0.9770 ± 0.0049
3, 4	0.9931 ± 0.0018	0.9697 ± 0.0143	0.9924 ± 0.0025	0.9770 ± 0.0049
5	0.9949 ± 0.0018	0.9697 ± 0.0143	0.9949 ± 0.0025	0.9770 ± 0.0049
6	0.9949 ± 0.0018	0.9647 ± 0.0135	0.9949 ± 0.0025	0.9754 ± 0.0085
7	0.9949 ± 0.0018	0.9647 ± 0.0135	0.9949 ± 0.0025	0.9754 ± 0.0085
8	0.9920 ± 0.0011	0.9647 ± 0.0135	0.9919 ± 0.0004	0.9754 ± 0.0085
9	0.9916 ± 0.0014	0.9647 ± 0.0135	0.9918 ± 0.0006	0.9754 ± 0.0085
10	0.9913 ± 0.0026	0.9647 ± 0.0135	0.9917 ± 0.0050	0.9754 ± 0.0085
11	0.9837 ± 0.0111	0.9837 ± 0.0111	0.9930 ± 0.0028	0.9930 ± 0.0028
12	0.9939 ± 0.0021	0.9705 ± 0.0076	0.9966 ± 0.0028	0.9814 ± 0.0036
13	0.9931 ± 0.0018	0.9539 ± 0.0105	0.9933 ± 0.0016	0.9727 ± 0.0036
14	0.9933 ± 0.0021	0.9705 ± 0.0076	0.9944 ± 0.0028	0.9814 ± 0.0036
15	0.9940 ± 0.0037	0.9940 ± 0.0037	0.9599 ± 0.0289	0.9599 ± 0.0289
16	0.9916 ± 0.0014	0.9398 ± 0.0249	0.9918 ± 0.0006	0.9612 ± 0.0167
17	0.9889 ± 0.0007	0.9889 ± 0.0007	0.9822 ± 0.0002	0.9822 ± 0.0002
18	0.9889 ± 0.0007	0.9889 ± 0.0007	0.9822 ± 0.0002	0.9822 ± 0.0002

Table 14: Emissivities and their uncertainties for the MSG3/4 split-window channels, per land cover type.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9968 ± 0.0022	0.9696 ± 0.0056	0.9973 ± 0.0026	0.9733 ± 0.0076
3, 4	0.9923 ± 0.0022	0.9696 ± 0.0056	0.9922 ± 0.0026	0.9733 ± 0.0076
5	0.9945 ± 0.0022	0.9696 ± 0.0056	0.9947 ± 0.0026	0.9733 ± 0.0076
6	0.9945 ± 0.0022	0.9680 ± 0.0120	0.9947 ± 0.0026	0.9726 ± 0.0092
7	0.9945 ± 0.0022	0.9680 ± 0.0120	0.9947 ± 0.0026	0.9726 ± 0.0092
8	0.9914 ± 0.0009	0.9680 ± 0.0120	0.9917 ± 0.0005	0.9726 ± 0.0092
9	0.9911 ± 0.0013	0.9680 ± 0.0120	0.9915 ± 0.0007	0.9726 ± 0.0092
10	0.9907 ± 0.0041	0.9680 ± 0.0120	0.9913 ± 0.0053	0.9726 ± 0.0092
11	0.9885 ± 0.0055	0.9885 ± 0.0055	0.9924 ± 0.0024	0.9924 ± 0.0024
12	0.9949 ± 0.0026	0.9728 ± 0.0052	0.9967 ± 0.0029	0.9780 ± 0.0031
13	0.9926 ± 0.0019	0.9574 ± 0.0066	0.9930 ± 0.0017	0.9711 ± 0.0037
14	0.9934 ± 0.0026	0.9728 ± 0.0052	0.9942 ± 0.0029	0.9780 ± 0.0031
15	0.9888 ± 0.0068	0.9888 ± 0.0068	0.9649 ± 0.0248	0.9649 ± 0.0248
16	0.9911 ± 0.0013	0.9469 ± 0.0211	0.9915 ± 0.0007	0.9658 ± 0.0068
17	0.9900 ± 0.0006	0.9900 ± 0.0006	0.9856 ± 0.0003	0.9856 ± 0.0003
18	0.9900 ± 0.0006	0.9900 ± 0.0006	0.9856 ± 0.0003	0.9856 ± 0.0003

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Table 15: Emissivities and their uncertainties for the Himawari-8/9 split-window channels, per land cover type.

IGBP Land Cover type	TIR1		TIR2	
	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$	$\epsilon_{veg} \pm \sigma_{\epsilon_{veg}}$	$\epsilon_{bs} \pm \sigma_{\epsilon_{bs}}$
1, 2	0.9967 ± 0.0020	0.9700 ± 0.0121	0.9975 ± 0.0025	0.9772 ± 0.0052
3, 4	0.9928 ± 0.0020	0.9700 ± 0.0121	0.9925 ± 0.0025	0.9772 ± 0.0052
5	0.9948 ± 0.0020	0.9700 ± 0.0121	0.9950 ± 0.0025	0.9772 ± 0.0052
6	0.9948 ± 0.0020	0.9657 ± 0.0130	0.9950 ± 0.0025	0.9756 ± 0.0088
7	0.9948 ± 0.0020	0.9657 ± 0.0130	0.9950 ± 0.0025	0.9756 ± 0.0088
8	0.9917 ± 0.0011	0.9657 ± 0.0130	0.9920 ± 0.0005	0.9756 ± 0.0088
9	0.9913 ± 0.0015	0.9657 ± 0.0130	0.9919 ± 0.0006	0.9756 ± 0.0088
10	0.9910 ± 0.0030	0.9657 ± 0.0130	0.9917 ± 0.0049	0.9756 ± 0.0088
11	0.9890 ± 0.0064	0.9890 ± 0.0064	0.9930 ± 0.0028	0.9930 ± 0.0028
12	0.9940 ± 0.0023	0.9712 ± 0.0068	0.9966 ± 0.0027	0.9814 ± 0.0040
13	0.9929 ± 0.0019	0.9558 ± 0.0094	0.9934 ± 0.0016	0.9721 ± 0.0039
14	0.9932 ± 0.0023	0.9712 ± 0.0068	0.9944 ± 0.0027	0.9814 ± 0.0040
15	0.9941 ± 0.0035	0.9941 ± 0.0035	0.9589 ± 0.0297	0.9589 ± 0.0297
16	0.9913 ± 0.0015	0.9415 ± 0.0243	0.9919 ± 0.0006	0.9588 ± 0.0238
17	0.9892 ± 0.0007	0.9892 ± 0.0007	0.9813 ± 0.0002	0.9813 ± 0.0002
18	0.9892 ± 0.0007	0.9892 ± 0.0007	0.9813 ± 0.0002	0.9813 ± 0.0002

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ANNEX 3: LST QUALITY CONTROL INFORMATION

Table 16: Bit-assignment in the LST quality flag.

Decimal value	Sat	Description
0	-	No data, Pixel out of Disk k
5	-	Sea
10	MSG (0° service)	Processed pixel, clear sky
15		Unprocessed pixel, cloud conditions
20	MSG IODC service	Processed pixel, clear sky
25		Unprocessed pixel, cloud conditions
30	Himawari	Processed pixel, clear sky
35		Unprocessed pixel, cloud conditions
40	GOES	Processed pixel, clear sky
45		Unprocessed pixel, cloud conditions
50	MULTI-MISSION	Processed pixel, clear sky
55		Unprocessed pixel, cloud conditions

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